

# Fake Job Detection Using Machine Learning

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**Abstract**—This project presents an end-to-end system for detecting fraudulent job postings using machine-learning techniques. The rise of digital recruitment has created opportunities for scammers to exploit job seekers, necessitating automated detection systems. The workflow begins with real-world data collection via web scraping from Internshala, followed by the integration of a labeled Kaggle dataset (EMSCAD) as the primary training source. The data undergoes extensive cleaning to address missing values and inconsistencies. To enhance predictive power, "meta-features" were engineered to capture structural patterns typical of scams, such as missing company logos or sparse descriptions. Textual content was consolidated and transformed using TF-IDF vectorization. Multiple algorithms, including Logistic Regression, Random Forest, and Support Vector Machine (SVM), were trained and evaluated, with a specific focus on addressing class imbalance using SMOTE. The SVM model emerged as the top performer. Finally, the optimized model was tested on newly scraped live job listings to demonstrate its effectiveness in real-world scenarios.

**Keywords**—machine learning, fraud detection, NLP, fake job postings, classification, SMOTE, TF-IDF

## I. Introduction

Online job platforms have become the standard for modern employment searching. However, this digital shift has inadvertently created a fertile environment for fraudulent job postings. Scammers leverage the anonymity and scale of the internet to publish deceptive listings designed to harvest personal information or solicit financial payments. These fake postings are often sophisticated, mimicking legitimate companies, which makes manual detection difficult and resource intensive.

Machine learning offers a robust solution by learning to recognize the textual and structural patterns that differentiate fraudulent postings from legitimate ones. However, building an effective detection system poses challenges, including severe class imbalance (far fewer fakes than real jobs) and unstructured text data.

This project aims to develop a complete machine-learning pipeline for detecting fake job postings. By combining real-world data acquisition with state-of-the-art NLP techniques and supervised learning, this project seeks to classify fraudulent postings accurately and provide a scalable methodology for safeguarding job seekers.

## II. Methodology

The technical process follows a complete pipeline from data collection to final validation.

### A. Data Acquisition & Sources

We utilized two distinct data sources. First, for training, we used the Kaggle "Real / Fake Job Posting Prediction" dataset (EMSCAD). It contains 17,880 entries with a binary label indicating whether a posting is real (0) or fake (1). Second, to test real-world applicability, we built a Python web scraper using requests and BeautifulSoup to collect live job postings from Internshala.com. These unlabeled entries were used for the final model demonstration.

Data successfully saved to 'scraped\_internshala\_jobs.csv'

--- Sample of Scraped Data ---

|   | Job Title                              | Company                     | Location                                    |
|---|----------------------------------------|-----------------------------|---------------------------------------------|
| 0 | Full Stack Development                 | She Can Foundation          | Work from home                              |
| 1 | Full Stack Development                 | SeaNeB Technologies Pvt Ltd | Anand                                       |
| 2 | Project Coordinator - AI Data Services | ConsultBae                  | Faridabad, Delhi, Ghaziabad, Gurgaon, Noida |
| 3 | Full Stack Development                 | GrowthGear                  | Faridabad, Delhi, Gurgaon, Noida            |
| 4 | UI/UX Testing/iOS Testing              | Across The Globe (ATG)      | Work from home                              |

Script finished.

Fig. 1. Sample of Scraped Data

### B. Exploratory Data Analysis (EDA)

EDA revealed a severe class imbalance in the training data: 17,014 entries (95.16%) were real, while only 866 (4.84%) were fraudulent. Additionally, critical fields such as 'salary\_range', 'company\_profile', and 'department' contained a high volume of missing values. Rather than discarding these rows, the sparsity

was identified as a potential indicator of fraud, as scammers often omit specific corporate details.

### C. Feature Engineering & Preprocessing

Text data in job descriptions and company profiles is often noisy. A standardized text cleaning function was implemented to normalize the text by converting it to lowercase, removing non-alphabet characters, and eliminating English stop words using the NLTK library.

Furthermore, an important part of this project was generating meaningful "meta-features" that capture structural differences between real and fake jobs. For each categorical field such as `has_company_logo` and `has_salary_range`, a binary indicator feature was created. This indicates whether the original field contains information (1) or is missing (0).

```

... --- 7. Starting Step 4: Creating Meta-Features ---
New binary 'meta-features' created.

--- Example of New Features ---
has_department  has_salary_range  has_company_profile  has_requirements \
0              1              0              1              1
1              1              0              1              1
2              0              0              1              1
3              1              0              1              1
4              0              0              1              1

has_benefits
0              0
1              1
2              0
3              1
4              1

--- Step 4 Complete ---

```

Fig. 2. Example of new Meta-Features

To convert processed text into numerical features, TF-IDF (Term Frequency–Inverse Document Frequency) vectorization was applied. TF-IDF measures how important each word or n-gram is within a specific job posting relative to its importance across the entire dataset.

## III. Machine Learning Modeling

The dataset was split into a Training Set (80%) and a Testing Set (20%). We trained and evaluated three distinct classifiers: Logistic Regression (as a baseline), Random Forest (an ensemble method), and Support Vector Machine (LinearSVC).

### A. Addressing Class Imbalance

Since fake job postings represent only about 5% of the dataset, class imbalance poses a major challenge. Without intervention, models tend to predict the majority class (real postings), resulting in poor recall for the fake class.

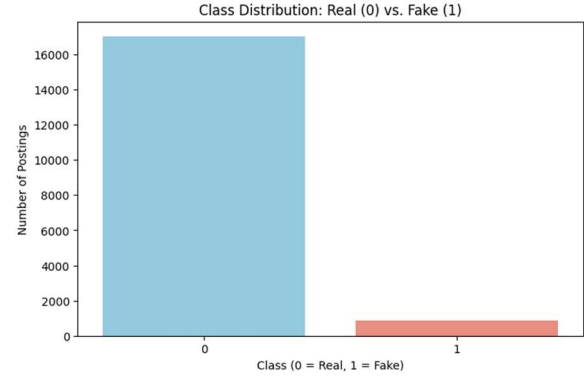


Fig. 3. Class distribution before SMOTE

To counter this, the Synthetic Minority Oversampling Technique (SMOTE) was applied after TF-IDF vectorization. SMOTE generates synthetic samples for the minority class by interpolating between existing minority examples, creating a balanced training environment.

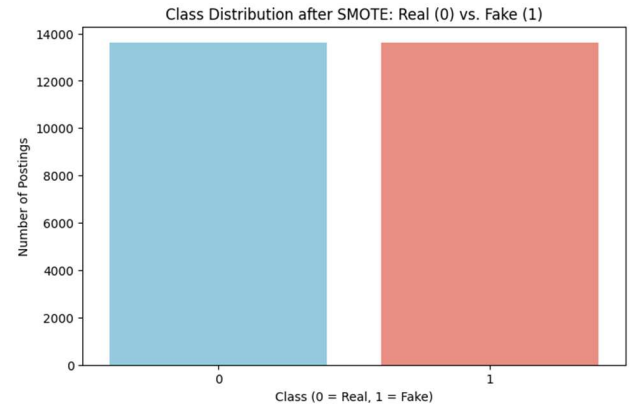


Fig. 3. Class distribution after SMOTE

### B. Evaluation Metrics

To assess model performance, we utilized Accuracy, Precision, Recall, and F1-Score. While Accuracy measures overall correctness, it can be misleading in imbalanced datasets. Therefore, we prioritized Precision (the percentage of flagged jobs that were actually fake) and Recall (the percentage of actual fake jobs that were correctly flagged). The F1-Score provides a harmonic mean of these two, offering a single metric to judge the model's effectiveness in identifying the minority class.

## IV. Results & Discussion

### A. Model Comparison

A comparative analysis revealed significant differences in performance. The results for the minority class (Fake Jobs) are summarized in Table I.

| Model                               | Precision   | Recall      | F1-Score    |
|-------------------------------------|-------------|-------------|-------------|
| Logistic Regression                 | 0.65        | 0.89        | 0.75        |
| Random Forest                       | 0.99        | 0.69        | 0.81        |
| <b>Support Vector Machine (SVM)</b> | <b>0.84</b> | <b>0.85</b> | <b>0.84</b> |

Table I. Model Results

### B. Discussion

Logistic Regression served as a strong baseline with high recall (0.89) but suffered from low precision (0.65), meaning it flagged many legitimate jobs as fake. Random Forest achieved near-perfect precision (0.99) but was too conservative, missing 31% of the scams (Recall 0.69).

The Support Vector Machine (SVM) model emerged as the top performer. It achieved the best balance, identifying 85% of all fake jobs while maintaining a high precision of 84%. The confusion matrix for the SVM model confirmed this strong performance, showing minimal misclassification for both classes.

### C. Final Validation

The optimized SVM model was applied to 50 live job postings scraped from Internshala. The model predicted "Real" for all postings, aligning with the platform's reputation. To validate detection capabilities, a synthetic "scam" posting was manually fed into the pipeline, which the model correctly classified as FAKE.

## V. Conclusions & Future Work

This project successfully developed a machine learning framework capable of detecting fake job postings with high accuracy. By combining NLP techniques (TF-IDF) with engineered meta-features and addressing class imbalance via SMOTE, the system effectively distinguishes between fraudulent and legitimate job ads. The Support Vector Machine classifier demonstrated the strongest generalization, making it a viable candidate for integration into job portals to protect job seekers.

Future work could focus on several key areas. First, implementing deep learning models such as BERT could capture more complex semantic patterns in job descriptions. Second, an advanced "deep scraper" could be developed to visit individual job pages and extract richer data, such as full benefit lists, rather

than relying on summary views. Finally, integrating explainable AI tools (like LIME or SHAP) would help users understand why a specific posting was flagged, increasing trust in the automated system.

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